

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester-III Examination, 2022-23

Botany (Industrial Microbiology)

Paper: IMB-301 — Agriculture and Food Microbiology

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions in all, selecting two each from Sections A and B and Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer the following in not more than 50 words each or fill in the blank:
[1 × 10 = 10]

- (a) What is commensalism?
- (b) Define siderophore.
- (c) What is the significance of nematode trapping fungi?
- (d) What is black gold?
- (e) Why is the red earthworm preferred for vermicomposting?
- (f) Define pasteurization.
- (g) Define water-activity.
- (h) What is Turkey-X-disease?
- (i) Botulism disease is caused by _____
- (j) Name the fungi used as single-cell protein.

SECTION A

Question 2. What are phosphate solubilizing micro-organisms? Describe the mechanism of solubilization of organic and inorganic phosphate by micro-organisms. [15]

Question 3. Describe briefly any two of the following: [7.5 × 2 = 15]

- (a) Role of symbiotic cyanobacteria in agriculture
- (b) VAM/AM fungi
- (c) Biological control of weeds

Question 4. Define biofertilizer and discuss briefly the advantages of liquid biofertilizers over carrier-based fertilizers. [15]

Question 5. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Ectomycorrhiza
- (b) Free-living nitrogen-fixing microbes
- (c) Suppressive and conducive soil

SECTION B

Question 6. Describe briefly the principles of food preservation and write an account of chemicals used for food preservation. [15]

Question 7. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Single-cell protein
- (b) Butter production
- (c) Nutritional properties of mushroom

Question 8. Give a comprehensive account of the role of intrinsic factors in microbial food spoilage. [15]

Question 9. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) Fermented bakery products
- (b) Food intoxication
- (c) Microbial spoilage of fish

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